HACKORBIT 2K26

Placement
Skill-Gap Hub

Final Technical Report

TEAM CODEAVENGERS

Track A — DataDrishti • BANNARI AMMAN INSTITUTE OF TECHNOLOGY

**From Placement Statistics
to Skill-Gap Intelligence.**

Sample / Demonstration Dataset — no real institutional student, placement, or recruiter data is used.

1. Executive Summary & Problem

One clear story from placement outcomes to training action

Placement Skill-Gap Hub is a Power BI decision-support solution that connects placement outcomes, industry skill demand, and student skill readiness into one actionable workflow. The core metric is the skill gap, which is classified by priority and translated into a plain-language training recommendation.

Official problem statement: “Placement Skill-Gap Hub — Model placement stats across branches.”

Traditional placement dashboards are useful for reporting placement rate, average package, companies, and branch performance. The gap is that they do not connect those outcomes to recruiter skill requirements or student readiness.

PLACEMENT OUTCOMES → INDUSTRY SKILL DEMAND → STUDENT SKILL READINESS → SKILL GAP → PRIORITY → TRAINING ACTION

Core question

Which skills should this institution prioritize to improve placement readiness?

The current MVP is deliberately scoped to a two-day build: a three-table demonstration dataset, three Power BI pages, core DAX measures, a Skill Gap Priority Matrix, a rule-based Training Recommendation Engine, and a branch × skill heatmap add-on.

2. Solution & Differentiation

Why this is more than a conventional placement dashboard

Conventional dashboard	Placement Skill-Gap Hub
Reports placement outcomes	Connects outcomes, demand, and readiness
Primarily descriptive	Adds diagnostic skill-gap analysis
Stops at charts	Ends with training priority
One major data layer	Three linked analytical layers

Defensible positioning: our differentiation is the integration of placement outcomes, recruiter skill demand, and student readiness into a single actionable skill-gap workflow. We do not claim this has never been built before.

Rule-based by design

The current recommendation engine is explicitly rule-based. It uses transparent thresholds and fixed recommendation templates; it is not AI or machine learning.

3. Dataset & Data Model

Small, explainable, hackathon-feasible architecture

Table	Key fields
Students_Placement	Student_ID, Branch, Batch, Placement_Status, Company, Package
Student_Skills	Student_ID, Skill, Assessment_Score (0–100)
Industry_Skill_Demand	Skill, Demand_Level (0–100), Companies_Requiring_Count

The model uses two active, single-direction relationships: Students_Placement[Student_ID] → Student_Skills[Student_ID], and Industry_Skill_Demand[Skill] → Student_Skills[Skill]. This keeps the model small and easy to debug and explain.

Demonstration scale: 240 students • 5 branches • 6 skills • 22 sample companies

Data transparency: Demand_Level and Assessment_Score use the same 0–100 scale for arithmetic comparison. They are treated as comparable proxies for a readiness gap, not as identical measurements.

4. Analytics Methodology & Core DAX

Transparent calculations that a judge can verify

Measure	Definition
Placement Rate	Students Placed ÷ Total Students
Average Package	Average package among placed students
Company Count	Distinct recruiting companies
Student Readiness %	Average Assessment_Score in the current skill/branch/batch context
Industry Demand %	Demand_Level for the selected skill
Skill Gap %	Industry Demand % – Student Readiness %
Skill Priority	CRITICAL ≥25 pts; HIGH ≥15; MEDIUM ≥5; otherwise LOW

Core formula: Skill Gap = Industry Demand – Student Readiness

Priority cut-points are deliberate, round-number starting thresholds rather than statistically derived boundaries. A real deployment would tune them against validated institutional outcome data.

5. Dashboard & Signature Features

Three pages, each answering one decision question

Dashboard page	Primary question	Core visuals
Placement Command Center	What is happening?	KPIs, branch placement, company hiring, package distribution
Skill Demand vs. Readiness	What does industry need, and are students ready?	Demand ranking, paired demand/readiness, gap table
Action Center	What should we do next?	Priority Matrix, Recommendation Engine, heatmap

Signature Feature 1 — Skill Gap Priority Matrix

Student Readiness is the X-axis and Industry Demand is the Y-axis. High demand + low readiness is the critical training quadrant.

Signature Feature 2 — Training Recommendation Engine

Each skill receives a gap, priority, and short rule-based recommendation. The completed Branch × Skill Heatmap localizes the largest gaps to a branch.

6. UX, Interactivity & Showcase

Modern presentation layer for the same analytical logic

The web showcase keeps the same data and core logic as the Power BI build while making the live demonstration easier to understand and operate.

- Interactive Branch, Batch, and Skill slicer buttons with clear active states.
- Unified filtering so branch/batch selections scope the skill-gap story as well as placement KPIs.
- Highest Priority Skill insight card that recalculates with the current filter context.
- Live Skill Gap formula display using the current top-priority skill.
- Branch Skill-Gap Heatmap for localized training priorities.
- Animated intro, modern micro-interactions, filter feedback, scroll progress, and quick evaluator demo path.
- Accessibility considerations including focus states and reduced-motion support.

15-second test: a first-time viewer should quickly identify placement health, the critical skill gap, the branch needing attention, and the next action.

7. Impact & Future Scope

Who benefits and what comes next

User	Value
Students	A ranked view of skills to build before placement season.
Placement / Training Team	A prioritized, evidence-based training plan.
Institution Leadership	A clearer basis for training budget and faculty time allocation.

Current MVP

- Three-table demonstration dataset
- Rule-based priority classification
- Manually curated skill list
- Static demonstration data
- Generic recommendation text by priority tier

Future scope

- Real institutional placement and recruiter-data integration
- ML-based placement prediction
- Automated skill extraction from resumes and job descriptions
- Continuous industry-demand updates
- Personalized student learning paths and training-impact simulation

Future capabilities are explicitly separated from what is working today.

8. Testing & Final Implementation Status

What was actually built, tested, and fixed

The final status records the dataset, all three Power BI pages, signature features, web dashboard, and testing as complete. Values were cross-checked against the raw dataset and the dashboard was exercised through real filter interactions.

Issue found	Fix
Web dashboard blank under restrictive networks	Removed external chart dependencies; visuals use local SVG rendering.
Placement Rate filter inconsistency	Explicit filter logic added so numerator and denominator use the intended context.
Skill Gap display scaling	Demand and readiness normalized before percentage formatting.
Industry Demand not changing with Skill	Corrected the field to the Industry_Skill_Demand source.
Heatmap layout width	Corrected markup so the heatmap matches the dashboard container.

Final verified build: 240 students • 5 branches • 6 skills • 22 sample companies • 3 Power BI pages • Priority Matrix • Recommendation Engine • Branch Skill-Gap Heatmap • web companion.

9. Judge Defense, Demo Flow & Conclusion

A compact final-defense sheet

Judge question	Answer
Where did the data come from?	Synthetic demonstration data; no real institutional data is claimed.
Is this AI?	No. The MVP uses a rule-based SWITCH / threshold engine.
How is the gap calculated?	Industry Demand – Student Readiness.
Does it prove causation?	No. It identifies actionable gaps; causal validation is future work.
How is it different?	It connects outcomes, demand, readiness, gap, and action in one workflow.
Can it scale?	Yes, by replacing the demonstration source with validated institutional data and integrations.

Recommended live demo

Placement Command Center → select Branch/Batch → Skill Demand vs. Readiness → show the largest gap → Action Center → show the priority and recommendation.

We don't just show placement numbers. We show what skills need to improve. Placement Skill-Gap Hub — From Placement Statistics to Skill-Gap Intelligence.

GitHub: codeavengers-placement-skill-gap • Vercel: codeavengers-placement-skill-gap.vercel.app

III.01

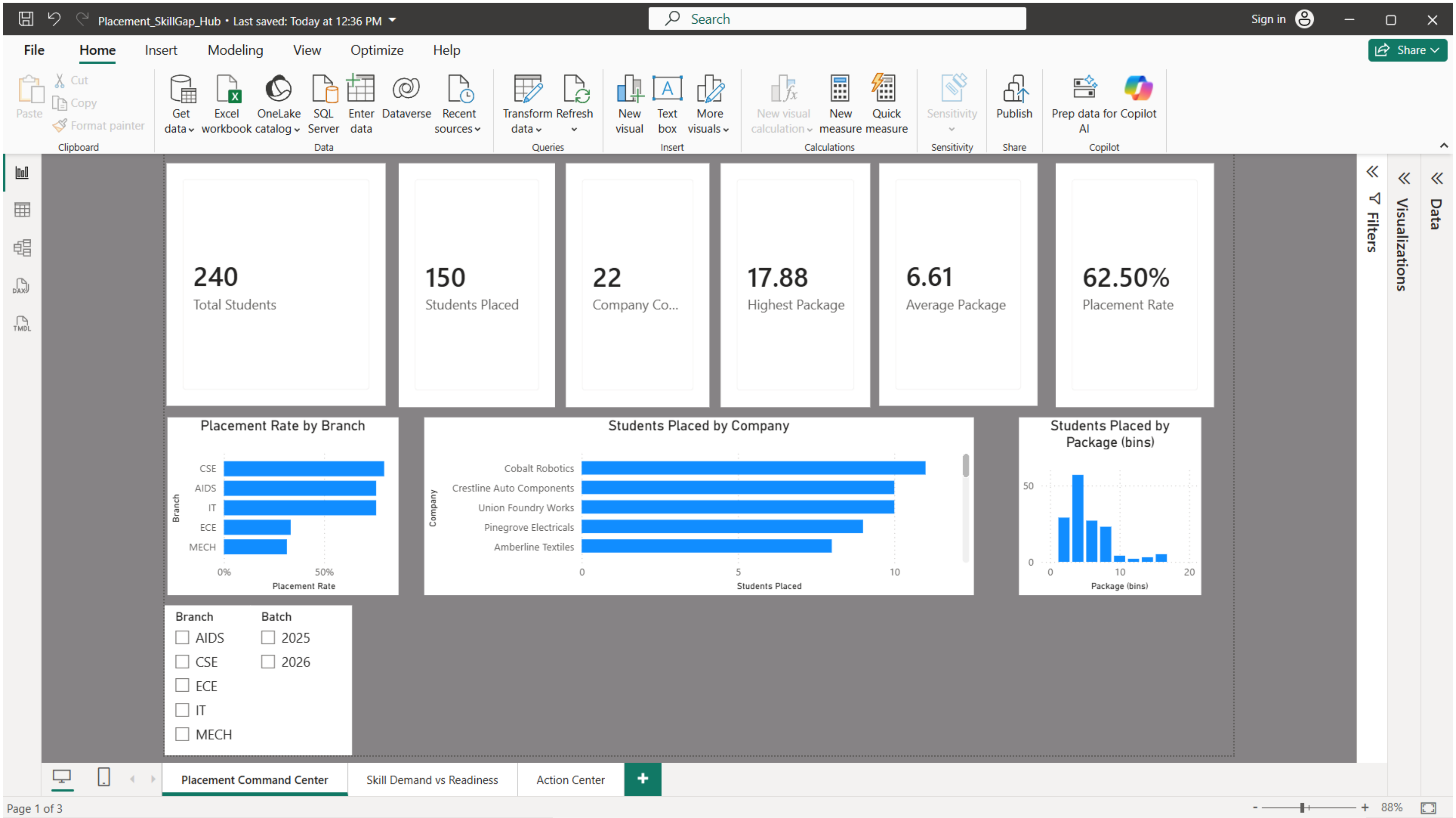
Power BI Desktop

The actual analytics implementation — DAX, data model, native slicers

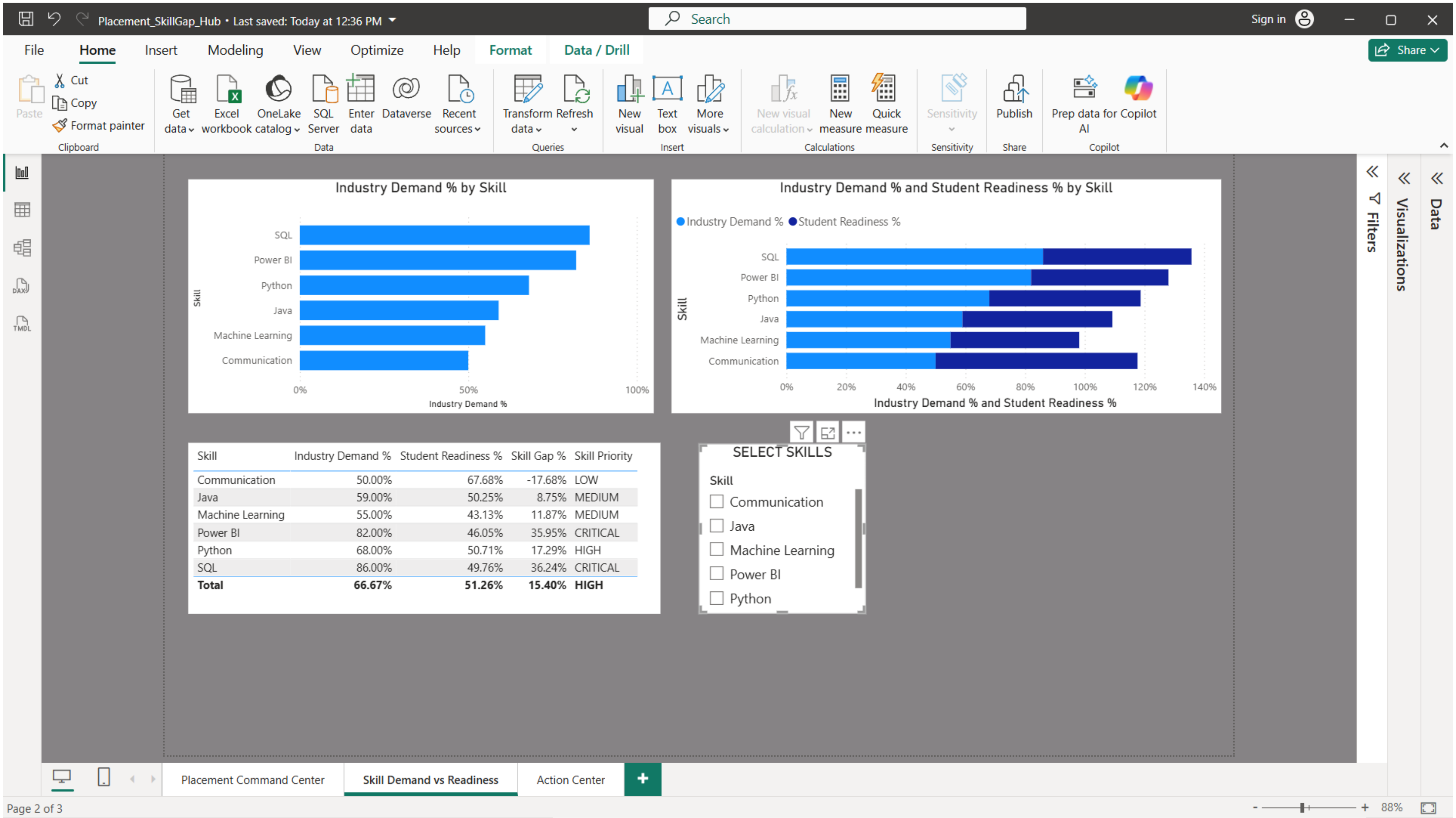
- Placement Command Center
- Skill Demand vs. Readiness
- Action Center — Priority Matrix & Recommendation Engine

POWER BI DESKTOP

Placement Command Center

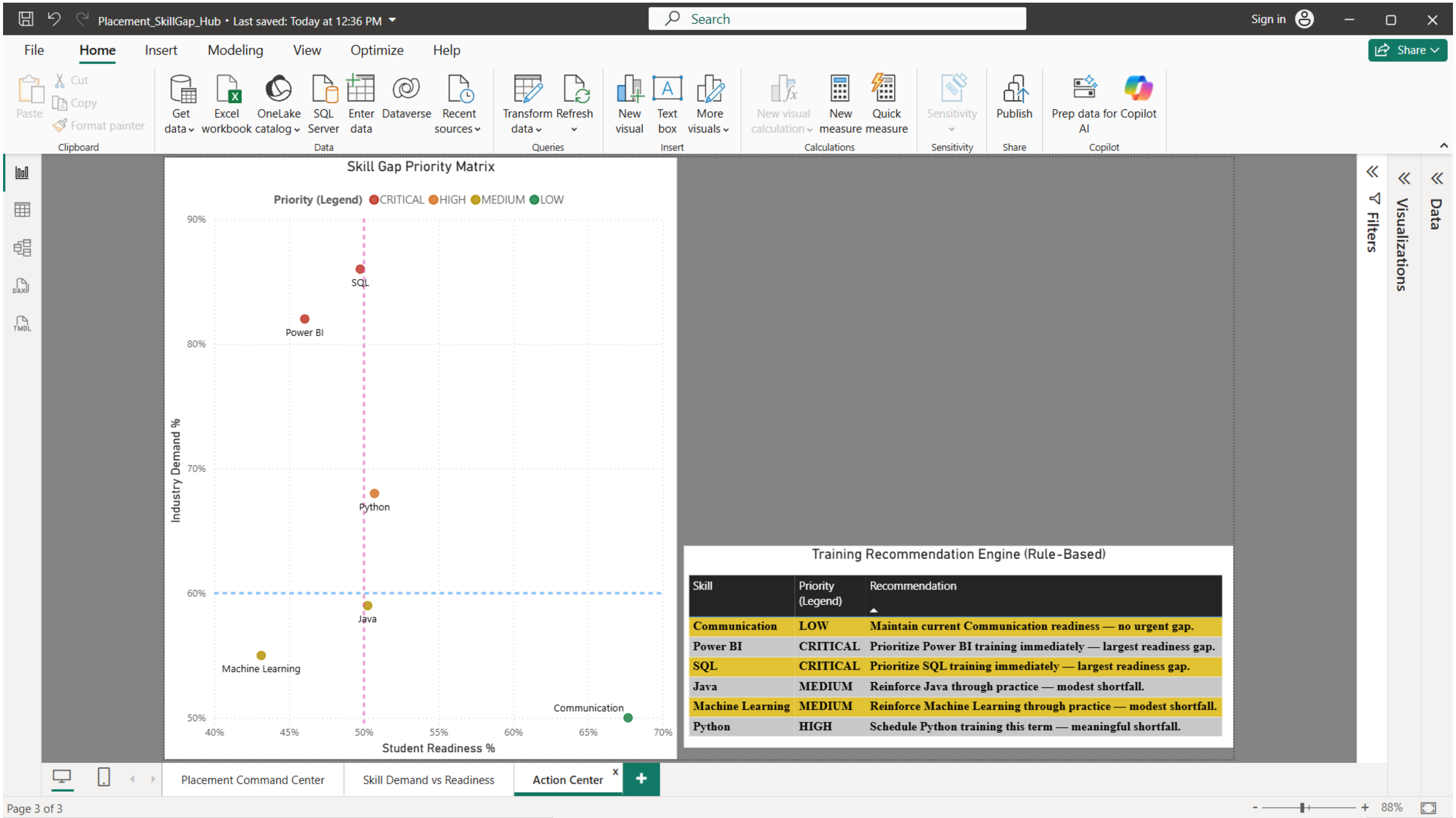


Skill Demand vs. Readiness



POWER BI DESKTOP

Action Center — Priority Matrix & Recommendation Engine



III.02

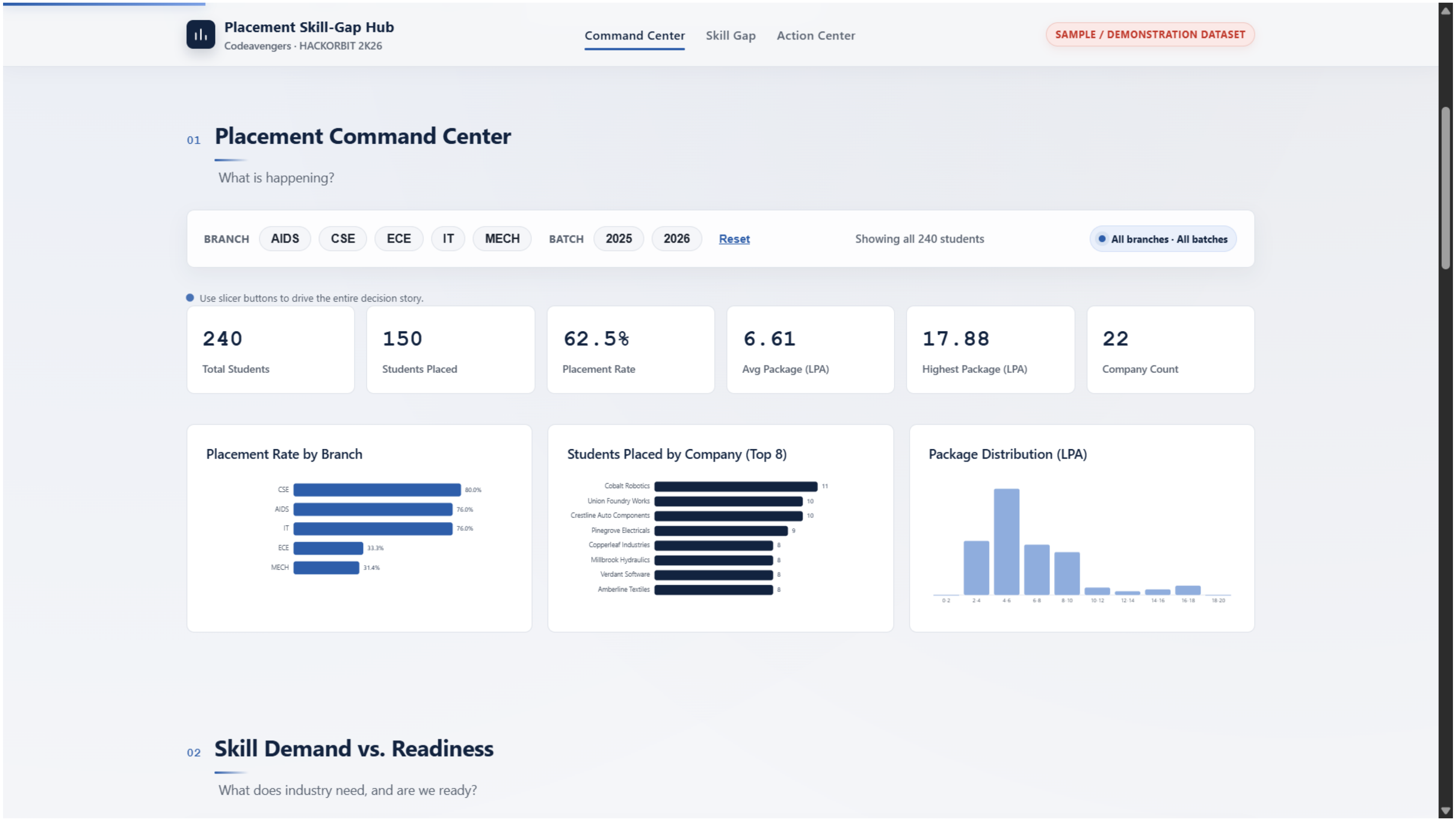
Web Dashboard

The live companion site — zero dependencies, same data and logic as Power BI

- Placement Command Center
- Skill Demand vs. Readiness
- Action Center — Priority Matrix & Recommendation Engine

WEB DASHBOARD — LIVE SITE

Placement Command Center



WEB DASHBOARD — LIVE SITE

Skill Demand vs. Readiness

Placement Skill-Gap Hub
Codeavengers · HACKORBIT 2K26

Command Center | Skill Gap | Action Center

SAMPLE / DEMONSTRATION DATASET

02 **Skill Demand vs. Readiness**

What does industry need, and are we ready?

SKILL Python SQL Power BI Machine Learning Java Communication Reset

CORE ANALYTIC

Skill Gap = Industry Demand − Student Readiness

`SQL 86.0% demand − 49.8% readiness = 36.2 percentage points`

A positive gap indicates a readiness shortfall relative to the demonstration demand level.

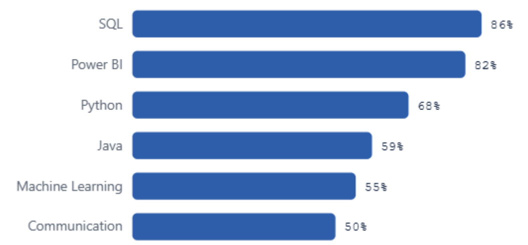
HIGHEST PRIORITY SKILL

SQL

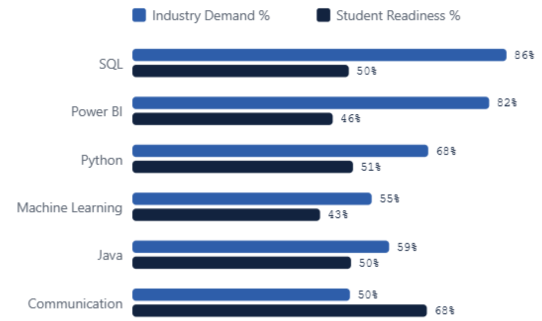
DEMAND	READINESS	GAP
86.0%	49.8%	36.2 pts

CRITICAL

Industry Demand % by Skill

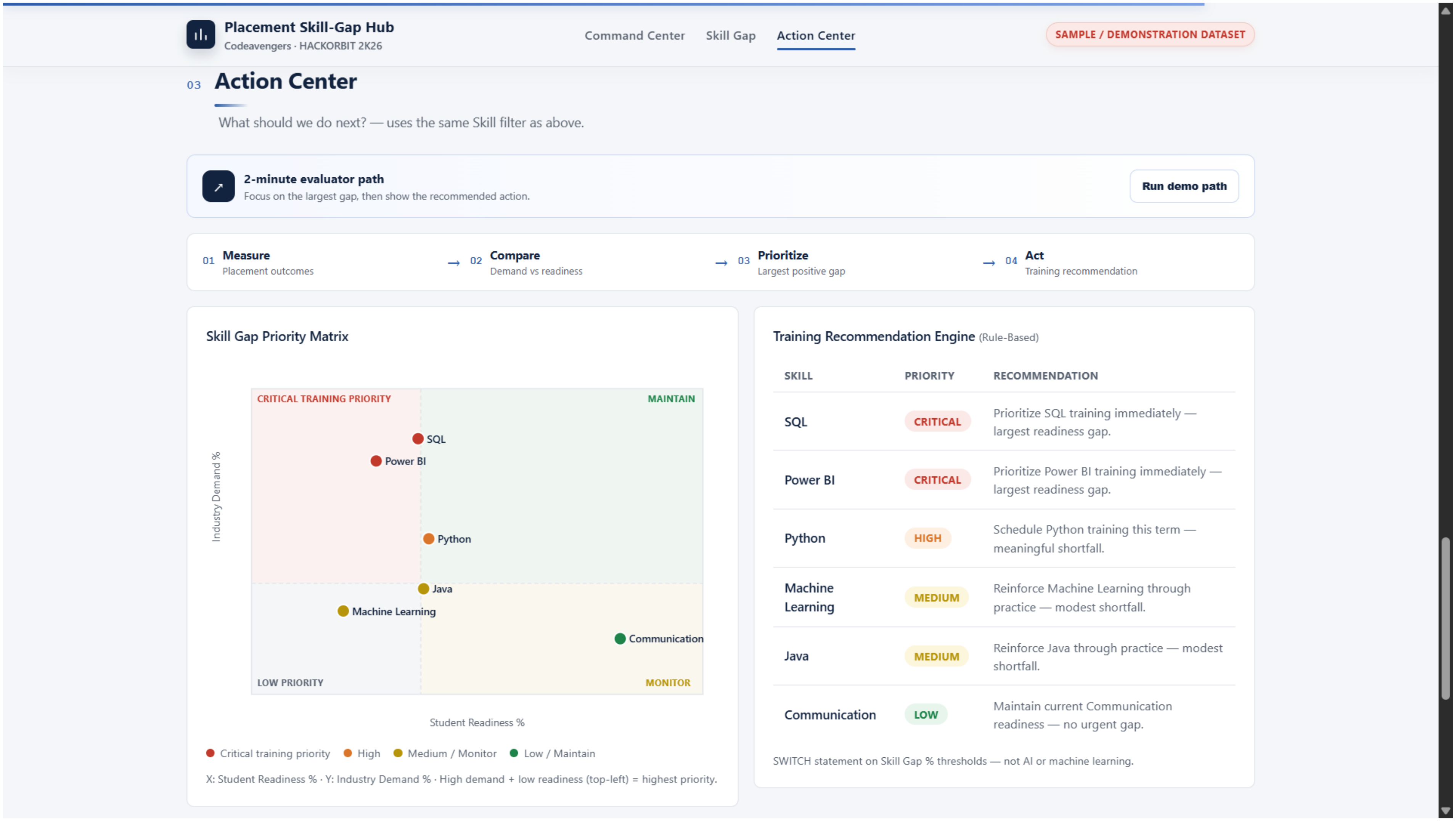


Demand % vs. Readiness % by Skill



WEB DASHBOARD — LIVE SITE

Action Center — Priority Matrix & Recommendation Engine



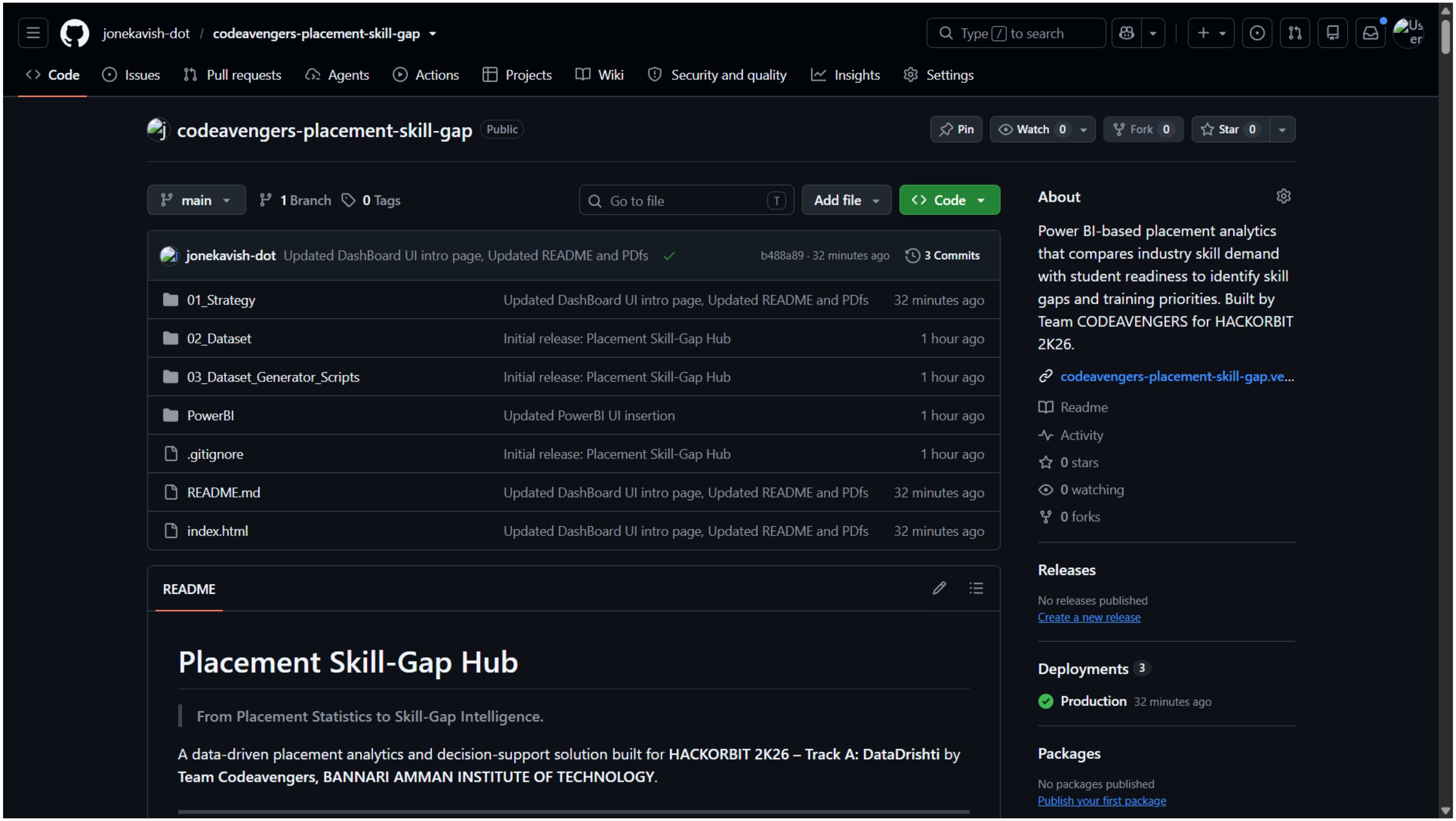
III.03

GitHub Repository

Full source: strategy docs, dataset, generator scripts, Power BI file, and web dashboard

- codeavengers-placement-skill-gap

Repository — codeavengers-placement-skill-gap



III.04

Vercel Deployment

The web dashboard, live and publicly reachable — no install required to view

- codeavengers-placement-skill-gap.vercel.app

VERCEL

Production Deployment